STEP-BY-STEP HOSTING GUIDE

Put PortaGraph Online for $0.

From the project folder on your computer to a live demo link your clients can open anywhere — using GitHub and Vercel, the same free setup behind your duct-designer demo. Written in plain English, every click spelled out.

~15 min	$0	3 steps
Total time, one coffee	GitHub + Vercel free tiers	Download, publish, deploy

BEFORE WE START

What you will end up with

Following this guide, you will publish PortaGraph exactly the way your **duct-designer-drag-drop** project is published: the code lives in a public **GitHub repository**, and the app runs on a free **Vercel** URL that anyone in the world can open — like `portagraph.vercel.app`. Nothing changes about the app itself; we are only giving it a public home.

Everything needed for this to work is already prepared inside the project: the database creates and fills itself on the first visit, the production build has been tested end-to-end, and no passwords, API keys, or settings are required anywhere in the process. If you can follow a recipe, you can do this.

STEP 1

Download & unzip

Get the project folder onto your computer.

STEP 2

Publish to GitHub

Upload the code to github.com/rajkdas.

STEP 3

Deploy on Vercel

Vercel builds it and gives you a live link.

0 PREPARE

What you need (2 minutes)

- ✓ **The project ZIP file** — `portagraph-mvp.zip`. Download it from this chat to your computer, then right-click it and choose **Extract All** (Windows) or double-click it (Mac). You should now have a folder named `portagraph-mvp` or `portagraph`.
- ✓ **A GitHub account** — you already have one (that is where `duct-designer-drag-drop` lives). Sign in at `github.com`.
- ✓ **A Vercel account** — go to `vercel.com` and click **Sign Up → Continue with GitHub**. It is one click, because Vercel simply borrows your GitHub login. No credit card is asked for.

Good to know: you do **not** need to install Node.js, a database, or any developer tools on your computer. The ZIP already contains tested, ready-to-run code — the computers at GitHub and Vercel do all the heavy work. Your laptop is only the courier.

1 GITHUB

Put the code on GitHub

GitHub is the online home of your code — think of it as Google Drive for software, with superpowers. Vercel (the next step) watches your GitHub repository, so this is the foundation that lets every future update deploy itself automatically.

Easy way — GitHub Desktop (recommended, no typing)

1. Install **GitHub Desktop** from `desktop.github.com`, open it, and sign in with your GitHub account.
2. Click **File → Add Local Repository...** and select your unzipped `portagraph` folder.
3. If it says the folder "is not a repository", click the button **Create a Repository Here**, leave everything at the defaults, and click "Create Repository" to confirm. (The ZIP usually already contains one, so you may not see this.)
4. Click **Publish repository** in the top bar.
5. In the dialog, keep the name `portagraph`, and **untick "Keep this code private"** so Vercel's one-click button works. Then click **Publish Repository**.
6. Wait for the upload to finish — the small progress wheel in the top-right corner tells you when it is done.

Alternative — Terminal (if you are comfortable with commands)

Open a terminal in the unzipped folder and paste these lines one at a time:

```
cd path/to/portagraph
git init
git add .
git commit -m "PortaGraph MVP"
git remote add origin https://github.com/rajkdas/portagraph.git
git branch -M main
git push -u origin main
```

When it asks for a login in the browser, approve it — that is GitHub confirming it is really you.

CHECKPOINT — Open `github.com/rajkdas/portagraph` in your browser. You should see the project files, and the front page should show a README with screenshots and a purple "Deploy with Vercel" button. If you can see that, step 1 is complete.

2 VERCEL

Deploy on Vercel (the magic button part)

Vercel is a free hosting service made by the same people who built Next.js — the technology PortaGraph is written in. It takes the code from your GitHub repository, builds it on their computers, and puts it online with HTTPS and a worldwide content network. This is exactly how your duct-designer demo is hosted.

1. Go to `vercel.com` and click **Sign In → Continue with GitHub**. Approve the access request if asked.
2. On your dashboard, click **Add New... → Project**.
3. Find **portagraph** in the "Import Git Repository" list and click **Import**. (If it is not listed, click **Adjust GitHub App Permissions → All Repositories** and retry.)
4. On the configuration screen, **change nothing**. See the box below for why every default is already correct.
5. Click **Deploy**. A build log appears — lines of text scroll for 2–3 minutes while Vercel compiles the app. This is normal.

6. When the log finishes, you get a celebration screen with a big **Continue to Dashboard** button. Your live web address is shown at the top — something like `portagraph.vercel.app` or `portagraph-xxxx.vercel.app`.

Why you change nothing: Vercel auto-detects Next.js and uses the right build command. PortaGraph needs **zero environment variables** — the SQLite database is created and filled with the 45-node demo network automatically on the first visit, at no extra setup. That is deliberate: this project was engineered so that "Import → Deploy" is genuinely the whole process.

CHECKPOINT — Click your new `.vercel.app` link. After a moment you should see the PortaGraph landing page — "Relationships that are portable, provable, payable." with a **Launch demo** button. That page is now publicly reachable from any phone or laptop on Earth.

3 VERIFY

Test the live demo (2 minutes)

Before showing it to anyone, walk the money path yourself — the exact route a client will take. Do these four clicks on your new link:

1. On the landing page click **Launch demo**, then choose **"Just exploring? Enter as the demo founder →"**. You land on the founder dashboard with the 45-member network loaded.
2. Open the **Matrix** tab and press **Simulate placement**. Watch the BFS animation walk the tree and the fee-split overlay appear — this is the moment that sells the product.
3. Open the **Admin** tab, find a HELD row in the Safety Holding panel, and press **Release**. The counter drops instantly — the 72-hour escrow demo.
4. Open the **Calculator** tab and drag any slider. Projections recalculate in real time — a nice closer for the pitch.

NOTE — The activity feed will show "polling indexer fallback (every 9s)" in its footer. That is correct and expected on Vercel; the realtime socket channel is a local-development extra, and the polling fallback keeps the feed fully alive.

4 POLISH

Point the README at your real URL (1 minute)

The README ships with `portagraph.vercel.app` as the demo link. If Vercel gave you a different address (for example `portagraph-abcd1234.vercel.app`), update it so visitors always click through to the right place:

1. Open your repository on `github.com` and click the pencil icon (✎) on `README.md`.
2. Replace the demo URL on the line near the top with your real one.
3. Click **Commit changes**. Vercel notices the new commit and redeploys automatically — you do not need to touch Vercel at all.

AFTER LAUNCH

Everyday use — the cheat sheet

I want to…	Do this
Change anything in the app	Edit the code, then push to GitHub (GitHub Desktop: write a summary → **Commit to main** → **Push origin**). Vercel redeploys automatically in 2–3 minutes.
Reset the demo data	Vercel dashboard → your project → **Deployments** → the ⋯ menu on the newest deployment → **Redeploy**. The database rebuilds itself fresh.
Use my own domain	Vercel → your project → **Settings → Domains** → **Add** (e.g. `demo.yourdomain.com`) and follow the two DNS instructions Vercel shows. HTTPS is automatic.
Check for errors	Vercel → your project → **Logs** (live runtime) or **Deployments → latest → Build Logs** (build time).
Undo a bad change	Vercel → **Deployments** → ⋯ on the last good deployment → **Instant Rollback**. Live again in seconds.

WHEN THINGS LOOK WRONG

Troubleshooting — plain-English answers

The Vercel build failed

Click **View Build Logs** and scroll to the first red lines. The most common cause is a removed `postinstall` script — `package.json` must contain `"postinstall": "prisma generate"`. It does in the shipped ZIP, so this only happens if scripts were edited. Copy the red lines into a chat with support (or back to me) and it is usually a one-line fix.

The site opens but data is missing

The very first request after a cold start creates and seeds the database, which takes 2–3 seconds — one slow first load is normal. If data never appears, open `/api/network` in a new tab: if that errors, check **Logs** in Vercel for the underlying message.

New sign-ups vanished after a while

Expected behaviour on the free tier: Vercel recycles idle servers, and each fresh server boots a clean, freshly seeded database. For a sales demo this is a feature — the demo always opens in its perfect 45-node state. Redeploy any time to force a reset.

The feed says "polling" instead of "streaming"

Correct on Vercel. True realtime push runs as a separate socket service for local development; hosted demos use the built-in 9-second polling fallback, which updates the feed continuously. Nothing is broken.

BONUS

Your 60-second demo script

When you show the site to a client, this order tells the story best — it is the same arc the product is built around:

1. **Landing page (10s):** "Every distributor's network becomes a soulbound asset they own — impossible to edit, delete, or steal."
2. **Matrix (20s):** press **Simulate placement** — "Placement is enforced by an immutable contract, and the expensive search runs off-chain: 87% gas savings."
3. **Admin (20s):** release a held payment — "Fees split instantly, but the pool share sits in a 72-hour safety window that a signer can reverse. Fraud control, on-chain."
4. **Calculator (10s):** drag a slider — "And this is what a member's projection looks like. All of this runs today, on a free tier."

REFERENCE

Where everything lives

Everything shipped in the ZIP — and now in your GitHub repository — has a job. Keep this map handy whenever you want to change something specific, and remember: any change you push to GitHub goes live automatically a couple of minutes later.

File / folder	What it is
README.md	The project's front page on GitHub — features, screenshots, API table and FAQ. Update the demo URL here after your first deploy.
DEPLOY.md	This same hosting guide in quick-reference form, committed to the repository so you never lose it.
src/	The application itself: landing page, matrix visualizer, founder dashboard, analytics, calculator and admin console.
contracts/	The real Solidity contracts — SBT registry, campaign manager, 72-hour escrow — plus their Foundry test suite.
prisma/	The data model for the off-chain mirror. The database creates and seeds itself from it on first run — no manual step.
docs/screenshots/	The four README screenshots. Swap them any time the interface changes to keep the repo looking fresh.

You're live. Ship it.

The demo now has the same publishing pipeline as your best repos: push to GitHub, and the world sees the update in minutes.

01 **Code lives at** github.com/rajkdas/portagraph — with README, screenshots, contracts and the MIT license.

02 **Demo runs at** your Vercel URL — auto-seeded database, no keys, no maintenance.

03 **Every future update** is one commit away: edit → commit → push → live. Rollbacks are instant.